

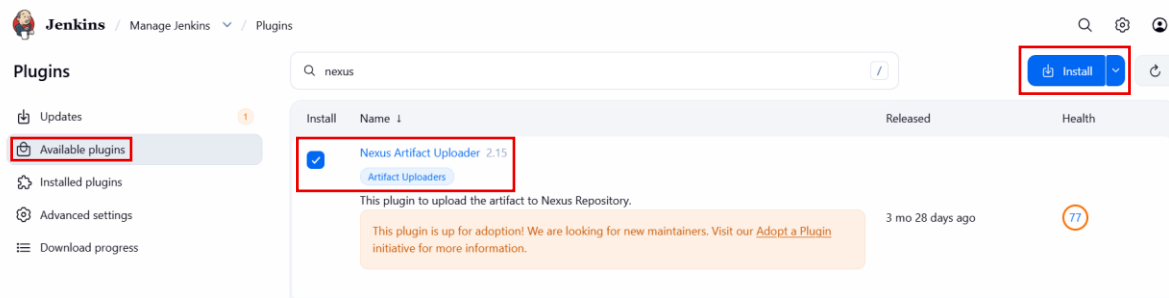
Nexus Working with Freestyle

To Begin with the Lab

Summary of the Lab

This lab demonstrates how to integrate **Jenkins Freestyle jobs with Nexus Repository** to upload build artifacts. First, install the **Nexus Artifact Uploader plugin** in Jenkins. Then create a **Freestyle project**, connect it to a **GitHub repository**, and configure the **Git source code**. Add a **Maven build step** with the goal package to generate a WAR file in the target/ directory. After the build, configure the **Nexus Artifact Uploader** as a post-build action, providing the Nexus URL, credentials, repository (maven-snapshots), and artifact details. Finally, run the build and verify that the **WAR artifact is successfully uploaded and stored in Nexus**.

- Go to Jenkins Dashboard
- Manage Jenkins → Manage Plugins
- Click:
 - Available Plugins
- Search for:
 - Nexus Artifact Uploader
- Select the plugin.
- Click:
 - Install Without Restart
- Wait for installation to complete.



- Create a Jenkins Freestyle Job
- Go to Jenkins Dashboard
- Click
 - New Item
- Enter job name
 - 25-nexus
- Select
 - Freestyle Project
- Click
 - OK

New Item

Enter an item name

25-nexus

Select an item type



Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

- Navigate Source Code Management
- Select the git option from there

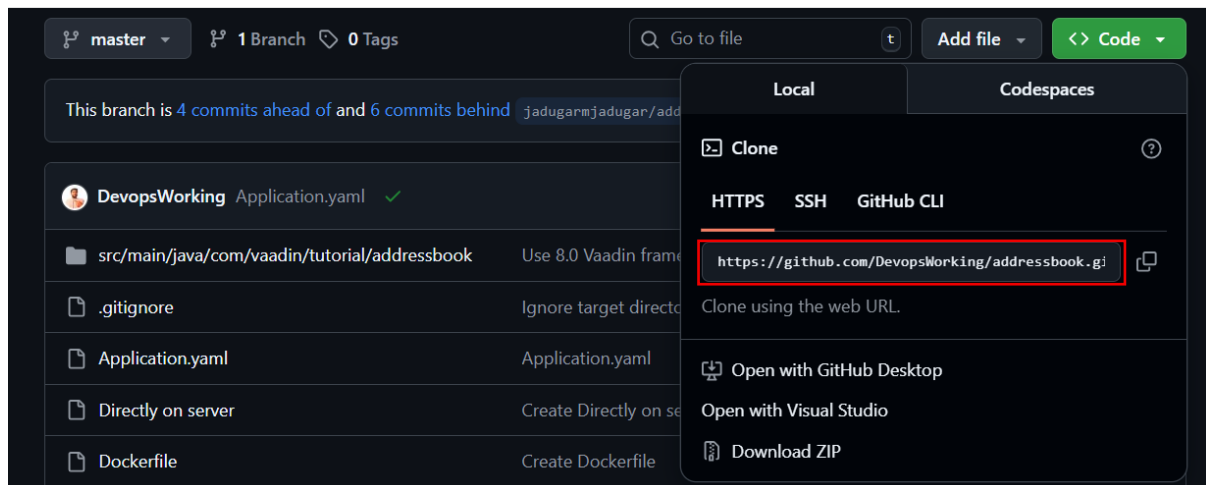
Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

☐ None

☒ Git ?

- After selecting git.
- Copy the URL from the github



- Enter the Repository URL
- Note: If your repository is private, you have to select the credentials from the credential option.

Repositories ?

Repository URL ?

`https://github.com/DevopsWorking/addressbook.git`

❗ Please enter Git repository.

Credentials ?

- none -

+ Add

Advanced ▾

- Check you github repository is master or main.

Branches to build ?

Branch Specifier (blank for 'any') ?

`*/master`

Environment

Configure settings and variables that define the context in which your build runs, like credentials, paths, and global parameters.

☒ Delete workspace before build starts

Advanced ▾

- ☐ Use secret text(s) or file(s) ?
- ☐ Send files or execute commands over SSH before the build starts ?
- ☐ Send files or execute commands over SSH after the build runs ?
- ☐ Add timestamps to the Console Output
- ☐ Inspect build log for published build scans

- Navigate to the Build Step section
- Click on the Add Build Step
- Select the 'Invoke top-level Maven targets'

Environment

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

+ Add build step

Filter

- Execute Windows batch command
- Execute shell
- Invoke Ant
- Invoke Gradle script
- Invoke top-level Maven targets**
- Run with timeout
- Set build status to "pending" on GitHub commit

... sending notifications, archiving artifacts, or triggering other jobs.

- Add Maven Build Step
- Scroll to Build Section
- Click
 - Add Build Step → Invoke top-level Maven targets
- Goals:
 - package
- This will generate the WAR file in:
- target/

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Invoke top-level Maven targets ?

Maven Version

Maven

Goals

package

Advanced

- Add Nexus Artifact Upload Step
- Click
 - Add Post-build Action → Nexus Artifact Uploader
- Configure the following:
 - Nexus Version
 - Nexus 3
 - Nexus URL
- Provide only the IP and Port.
- Example:
 - <EC2_PUBLIC_IP>:8081
- ⚠ Do NOT include http://

Nexus artifact uploader

Nexus Details

Nexus Version

NEXUS3

Protocol

HTTP

Nexus URL ?

44.198.178.55:8081/

- Credentials

- Click
- Add → Jenkins
- Enter:
 - Username: admin
 - Password: <nexus_admin_password>
 - ID: nexus
 - Description: nexus

The screenshot shows the 'Add Username with password' dialog in Jenkins. The fields are as follows:

- Scope:** Global (Jenkins, nodes, items, all child items, etc)
- Username:** admin
- Treat username as secret:** ☐
- Password:** (masked with dots)
- ID:** nexus (with a red error message: "This ID is already in use")
- Description:** nexus
- Create Button:** A blue button labeled 'Create'.

- Click
 - Add
 - Then select the credential.
- Configure Artifact Details
- Group ID
- Example:
 - Addressbook
- Version
- For snapshot builds:
 - 1.0-SNAPSHOT

- Explanation:
 - Release version → final production artifact
 - Snapshot version → development builds
- Repository
- Select:
 - maven-snapshots

Credentials

admin/***** (nexus) + Add

GroupId

addressbook

Version

1.0-SNAPSHOT

Repository ?

maven-snapshots

- Configure Artifact File Upload
- Click
 - Add Artifact
- Fill details:
 - Artifact ID:
 - Dev
 - Type:
 - war
 - Classifier:
 - (empty)
 - File Path:
 - target/addressbook-2.0.war

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions**

Artifacts

Artifact

ArtifactId

Dev

Type ?

war

Classifier ?

File ?

target/addressbook-2.0.war

- Click Save
- Run the Build
- Click

- Build Now
- Open
- Console Output

Uploaded: <http://44.198.178.55:8081/repository/maven-snapshots/addressbook/Dev/1.0-SNAPSHOT/Dev-1.0-20260314.181747-1.war> (14 MB at 16 MB/s)

Uploading: <http://44.198.178.55:8081/repository/maven-snapshots/addressbook/Dev/1.0-SNAPSHOT/maven-metadata.xml>

Uploaded: <http://44.198.178.55:8081/repository/maven-snapshots/addressbook/Dev/1.0-SNAPSHOT/maven-metadata.xml> (587 B at 1.9 kB/s)

Uploading artifact addressbook-2.0.war completed.
Finished: SUCCESS

- Verify Artifact in Nexus
- Open Nexus:
 - http://<EC2_PUBLIC_IP>:8081
- Go to:
 - Browse → maven-snapshots
- Navigate:
 - Group ID → Artifact → Version
- Example:
 - addressbook
 - └─ addressbook
 - └─ 1.0-SNAPSHOT
- Your WAR file will be stored here.

